

# A power lower bound for every permutation-fair die

Research proof: independently reviewed

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**Status.** Complete proof with independent mathematical reviews by the three other research agents. These are not formal proof-assistant checks. Its essential input is the finite mean-deficit balance theorem in `rational_designs/asymptotic/finite_endpoint_odds_balance.tex`.

**Theorem 1.** *There is an absolute constant  $c > 0$  such that every individual die in a permutation-fair collection of  $n$  equiprobable finite dice has at least  $cn^{3/2}$  faces. More generally, suppose  $p_1, \dots, p_m : [K] \rightarrow [0, 1]$  are nondecreasing, the rows have positive probability weights  $w_q$ , and*

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]).$$

*Then  $w_1, w_K \leq Cm^{-3/2}$  for an absolute  $C$ .*

All constants denoted by  $C, c$  below are absolute and may change from line to line. Fixed numerical choices, such as  $\theta = 1/16$ , do not depend on  $m$  or on the representation. We prove the assertion for sufficiently large  $m$ ; increasing  $C$  covers the remaining range.

## The finite balance input

Put  $u_j = 1 - p_j$  and  $s = \sum_j u_j$ . The cited finite balance theorem and its row-mass corollary give, for  $m \geq 256$ ,

$$\max_j u_j \leq 256(s + 1)/m, \tag{1}$$

$$\Pr_w(s \leq S) \leq 512(S + 1)/m \quad (S \geq 0). \tag{2}$$

The first bound is the cited theorem when  $s \leq m/256$ ; otherwise it follows simply from  $u_j \leq 1$ . In particular, for every fixed  $a > 0$ ,

$$\mathbb{E}_w e^{-as} \leq C_a/m. \tag{3}$$

We shall also use the following uniform consequence for integers  $k \geq 0$ :

$$\mathbb{E}_w (s + 1)^k e^{-as} \leq \frac{C_a^{k+1} k!}{m}. \tag{4}$$

Indeed split  $e^{-as} = e^{-as/2} e^{-as/2}$ , apply (3), and use  $\sup_{s \geq 0} (s + 1)^k e^{-as/2} \leq C_a^k k!$ .

## A stable polarization comparison

For a polynomial  $P(z) = \sum_k a_k z^k$  of degree at most  $N$ , write

$$\text{Pol}_N(P)(z_1, \dots, z_N) = \sum_k a_k \frac{e_k(z_1, \dots, z_N)}{\binom{N}{k}}.$$

This is its symmetric multiaffine polarization.

**Lemma 1** (Centered elementary coefficients). *If  $\sum_{i=1}^N y_i = 0$  and  $\sigma^2 = \sum_i y_i^2$ , then for  $1 \leq k \leq N$ ,*

$$\frac{|e_k(y)|}{\binom{N}{k}} \leq \left( \frac{\sqrt{ek} \sigma}{N} \right)^k.$$

*Proof.* The case  $\sigma = 0$  is immediate. The elementary inequality  $|1 + u|^2 \leq \exp(2 \operatorname{Re} u + |u|^2)$  holds for every complex  $u$ . Since the linear terms sum to zero, it gives

$$\left| \prod_i (1 + zy_i) \right| \leq \exp(\sigma^2 |z|^2 / 2)$$

for every complex  $z$ . Cauchy's coefficient bound at radius  $\sqrt{k}/\sigma$  gives  $|e_k| \leq (\sqrt{e} \sigma / \sqrt{k})^k$ . Divide by  $\binom{N}{k} \geq (N/k)^k$ .  $\square$

**Lemma 2** (A variance-adapted Laguerre derivative bound). *For integers  $j \geq h \geq 0$  and  $z > 0$ ,*

$$z^{h/2} \left| \frac{d^h}{dz^h} L_j(\theta z) \right| \leq (\theta j)^{h/2} e^{\theta z/2}.$$

*For  $h > j$  the derivative is zero.*

*Proof.* Give entire functions the Gaussian norm  $\|f\|^2 = \pi^{-1} \int_{\mathbb{C}} |f(w)|^2 e^{-|w|^2} dA(w)$ . The functions  $f_j(w) = w^j / \sqrt{j!}$  are orthonormal. The displacement

$$(D_a f)(w) = e^{-|a|^2/2 + aw} f(w - \bar{a})$$

is an isometry, by the change of variables  $w = u + \bar{a}$ . For real  $a = \sqrt{y}$ , direct coefficient expansion gives

$$|\langle f_{j-h}, D_a f_j \rangle| = e^{-y/2} \sqrt{\frac{(j-h)!}{j!}} y^{h/2} \left| L_{j-h}^{(h)}(y) \right| \leq 1.$$

Here the superscript in  $L_{j-h}^{(h)}$  is its associated parameter; indeed the coefficient of  $w^{j-h}$  in  $e^{aw}(w-a)^j$  is  $(-a)^h L_{j-h}^{(h)}(a^2)$ . The differentiation identity  $\frac{d^h}{dy^h} L_j(y) = (-1)^h L_{j-h}^{(h)}(y)$  and  $j!/(j-h)! \leq j^h$  prove the unscaled assertion. Apply the chain rule with  $y = \theta z$ .  $\square$

Fix from now on

$$\ell = \lfloor m/4 \rfloor, \quad N = m - \ell, \quad \alpha = \ell/m, \quad \theta = 1/16.$$

Let  $B$  be a uniformly chosen  $\ell$ -element subset of  $[m]$ , and put

$$A_B = \prod_{j \in B} p_j, \quad \bar{A} = \mathbb{E}_B A_B, \quad x = \alpha s.$$

The measure  $\mu$  on rows defined by

$$\mu(q) = (\ell + 1)w_q \bar{A}(q)$$

is a probability measure. Define also the probability density

$$b_\ell(z) = \frac{\ell + 1}{\ell} (1 - z/\ell)^\ell \mathbf{1}_{[0, \ell]}(z).$$

Its expectation functional is denoted by  $\beta_\ell$ .

Let  $L_j$  be the ordinary Laguerre polynomials, orthonormal for  $e^{-z} dz$  on  $[0, \infty)$ .

**Lemma 3** (Comparison with a scalar endpoint measure). *Let  $1 \leq d \leq \sqrt{m}$  and*

$$Q_a(z) = \sum_{j=0}^d c_{aj} L_j(\theta z), \quad B_a = \sum_j |c_{aj}| \quad (a = 1, 2).$$

For  $P = Q_1 Q_2$ , one has

$$|\mathbb{E}_\mu P(x) - \beta_\ell(P)| \leq C B_1 B_2 \left( \frac{d}{m} + e^{-cm} \right). \quad (*)$$

The constants are uniform in the coefficients.

*Proof.* Lemma 2 and the product rule give, for  $z > 0$ ,

$$\frac{|P^{(k)}(z)|}{k!} \leq B_1 B_2 e^{\theta z} \frac{(2\sqrt{\theta d/z})^k}{k!}. \quad (5)$$

We will occasionally also use the coarser, nonsingular derivative bounds  $|P'(z)| \leq C B_1 B_2 d e^{\theta z}$  and  $|P(z)| \leq B_1 B_2 e^{\theta z}$ . The former follows from  $|L'_j(y)| \leq j e^{y/2}$ , using  $L'_j = -\sum_{v < j} L_v$  and the  $h = 0$  case of Lemma 2.

For fixed  $B$ , put  $z_j = \ell u_j$  for  $j \notin B$ , and let  $v_B = N^{-1} \sum_{j \notin B} z_j$  be their mean. Multilinearity and the exact mixed moments give

$$(\ell + 1) \mathbb{E}_w \mathbb{E}_B [A_B \text{Pol}_N(P)(z_{[m] \setminus B})] = \beta_\ell(P). \quad (6)$$

Indeed, after expansion every distinct-coordinate product can be replaced by the corresponding product of one common uniform variable  $t$ . The right side becomes  $(\ell + 1) \int_0^1 t^\ell P(\ell(1-t)) dt$ . The degree condition holds because  $\deg P \leq 2d \leq N$  for large  $m$ .

On every row, (1) gives  $\max_j z_j \leq C(s + 1)$ , while

$$0 \leq x, v_B \leq \frac{\alpha}{1 - \alpha} s, \quad \sigma_B^2 := \sum_{j \notin B} (z_j - v_B)^2 \leq \sum_{j \notin B} z_j^2 \leq C N (s + 1) v_B.$$

Maclaurin's inequality gives

$$\bar{A} = \frac{e\ell(p)}{\binom{m}{\ell}} \leq (1 - s/m)^\ell \leq e^{-\alpha s}. \quad (7)$$

If  $v_B = 0$ , all its nonnegative  $z_j$  are zero and polarization has no error. Otherwise Taylor expansion about  $v_B$  is exact:

$$\text{Pol}_N(P)(z) - P(v_B) = \sum_{k=2}^{\deg P} \frac{P^{(k)}(v_B)}{k!} \frac{e_k(z - v_B)}{\binom{N}{k}}.$$

The factors  $v_B^{-k/2}$  in (5) cancel the factors  $v_B^{k/2}$  in Lemma 1. Thus the absolute difference is at most

$$B_1 B_2 e^{\theta v_B} \sum_{k=2}^{2d} \frac{(C \sqrt{dk(s+1)/m})^k}{k!}.$$

As  $\theta\alpha/(1-\alpha) < \alpha/4$ ,  $\mathbb{E}_B A_B e^{\theta v_B} \leq e^{-3\alpha s/4}$ . Using (4), with  $\lceil k/2 \rceil$  in place of  $k$ , and multiplying by  $\ell + 1$ , bounds this part of the integrated error by

$$C B_1 B_2 \sum_{k=2}^{2d} (C \sqrt{d/m})^k \frac{k^{k/2} \lceil k/2 \rceil!}{k!}.$$

The last ratio is at most  $C^k(k+1)$ , as follows directly from  $k! \geq (k/e)^k$  and  $\lceil k/2 \rceil! \leq \lceil k/2 \rceil^{k/2}$ . For large  $m$  and  $d \leq \sqrt{m}$ , the resulting geometric series is at most  $Cd/m$ . The polarization error is therefore  $C B_1 B_2 d/m$ .

It remains to replace  $P(v_B)$  by  $P(x)$ . Restrict first to  $s \leq m/256$ , where all  $p_j \geq 1/2$ . Under the success-weighted law  $\Pr(B) = A_B/e_\ell(p)$ , put  $\Delta = v_B - x$ . The exact bias identity is

$$\mathbb{E}\Delta = \frac{\ell}{mN} \sum_{i < j} (p_i - p_j)^2 \frac{e_{\ell-1}(p_{-[i,j]})}{e_\ell(p)}.$$

The ratio is at most  $1/\min p_i \leq 2$ . Moreover the subset-inclusion covariances are nonpositive by Newton's inequality, giving  $\text{Var } \Delta \leq N^{-2} \sum_j (\ell u_j)^2$ . These facts are proved in detail in `rational_designs/asymptotic/weighted_success_subset_moments.tex`. Since

$$\sum_j u_j^2 \leq s \max_j u_j \leq C s(s+1)/m,$$

they give the sharper bounds

$$0 \leq \mathbb{E}\Delta \leq C s(s+1)/m, \quad \mathbb{E}\Delta^2 \leq C(s(s+1)/m + s^2(s+1)^2/m^2). \quad (8)$$

If  $s = 0$ , all the coordinates vanish and there is no averaging error. Otherwise  $x = \alpha s > 0$ . Taylor's exact integral remainder and (5) give

$$|P(v_B) - P(x) - P'(x)\Delta| \leq C B_1 B_2 d e^{\theta \max(x, v_B)} \frac{\Delta^2}{x}.$$

Indeed  $x + t\Delta = (1-t)x + tv_B \geq (1-t)x$ , so  $(1-t)/(x+t\Delta) \leq 1/x$  in the remainder integral. This also handles  $v_B = 0$  by a limit; there is no singular endpoint error. Combining with (8) and the first derivative estimate in (5), the weighted averaging error is at most

$$C B_1 B_2 e^{\theta\alpha s/(1-\alpha)} \left[ \frac{\sqrt{d} \sqrt{s(s+1)}}{m} + \frac{d(s+1)}{m} + \frac{ds(s+1)^2}{m^2} \right].$$

Multiply by  $\bar{A}$ , use (7) and (4), and then multiply by  $\ell + 1$ . Its contribution is at most  $C B_1 B_2 d/m$ .

For rows with  $s > m/256$ , a coarser estimate suffices. Under a uniformly chosen subset,  $\mathbb{E}_B v_B = x$  and

$$\mathbb{E}_B (v_B - x)^2 = \frac{\ell}{N(m-1)} \frac{1}{m} \sum_{j=1}^m (\ell u_j - x)^2 \leq C(s+1)^2/m.$$

Since  $0 \leq A_B \leq 1$ , Cauchy-Schwarz gives

$$\mathbb{E}_B A_B |v_B - x| \leq e^{-\alpha s/2} C(s+1)/\sqrt{m}.$$

The nonsingular first-derivative bound is  $C B_1 B_2 d e^{\theta\alpha s/(1-\alpha)}$  on the relevant interval. As  $m/256 < s \leq m$  and  $d \leq \sqrt{m}$ , the integrated error, even including the factor  $\ell + 1$ , is at most  $C B_1 B_2 e^{-cm}$ . Together with (6), this proves (\*).  $\square$

## Endpoint tests with an exponential margin

For  $d \geq 1$  define

$$T_d(y) = \sum_{j=0}^{d-1} (1 - j/d) L_j(y), \quad R_d(z) = \left( \frac{2}{\theta d} - z \right) T_d(\theta z)^2.$$

Laguerre orthogonality and the three-term recurrence give

$$\int_0^\infty e^{-y} T_d(y)^2 dy = \frac{(d+1)(2d+1)}{6d}, \quad \int_0^\infty y e^{-y} T_d(y)^2 dy = \frac{d+1}{2d}.$$

Thus the mean of  $z$  under  $e^{-\theta z} T_d(\theta z)^2 dz$  is  $3/[\theta(2d+1)] \leq 3/(2\theta d)$ . The density ratio  $b_\ell(z)/e^{-\theta z}$  is nonincreasing, including its drop to zero beyond  $\ell$ . Multiplication by this ratio cannot increase the mean: this follows directly by integrating  $(z - z')(r(z) - r(z')) \leq 0$  against two copies of the original measure. Consequently

$$\beta_\ell(R_d) \geq \frac{1}{2\theta d} \beta_\ell(T_d(\theta z)^2).$$

For  $0 \leq z \leq 1/(4d)$  and  $j < d$ ,  $|L_j(\theta z) - 1| \leq e^{j\theta z} - 1 < 1/3$ . Hence  $T_d(\theta z) \geq d/3$  there. Moreover  $b_\ell(z) \geq 3/4$  on that interval, by  $(1 - z/\ell)^\ell \geq 1 - z$ . It follows that

$$\beta_\ell(R_d) \geq \frac{1}{96\theta}. \tag{9}$$

The Laguerre recurrence also gives the useful exact identity

$$y T_d(y) = \frac{1}{d} \sum_{j=0}^{d-1} L_j(y) - L_d(y).$$

Consequently  $R_d = T_d(\theta z) S_d(z)$ , where

$$S_d(z) = \frac{1}{\theta d} \sum_{j=0}^{d-1} (1 - 2j/d) L_j(\theta z) + \frac{1}{\theta} L_d(\theta z).$$

The sums of absolute Laguerre coefficients are at most  $d$  for  $T_d$  and  $2/\theta$  for  $S_d$ . Choose  $d = \lfloor \kappa \sqrt{m} \rfloor$ , with a sufficiently small absolute  $0 < \kappa \leq 1$ . Lemma 3 bounds the error for  $R_d$  by

$$C \left( \frac{d^2}{m} + d e^{-cm} \right).$$

By first choosing  $\kappa$  small and then  $m$  large, this is smaller than half the positive constant in (9). Thus  $\mathbb{E}_\mu R_d(x) > 0$ . Since  $R_d$  is nonpositive for  $x \geq 2/(\theta d)$ , some row of positive  $\mu$ -mass lies below that threshold. The last row minimizes  $s$  and  $x$ , so

$$x_K < 2/(\theta d). \tag{10}$$

Put  $r = \lfloor d/8 \rfloor$  and

$$S = \sum_{j=0}^r L_j(\theta x_K)^2, \quad Q_*(z) = \frac{1}{S} \sum_{j=0}^r L_j(\theta x_K) L_j(\theta z).$$

Then  $Q_*(x_K) = 1$ . From (10) we have  $j\theta x_K \leq 1/4$ , so  $|L_j(\theta x_K) - 1| < 1/3$ . In particular  $S \geq cd$ , and the sum of absolute Laguerre coefficients of  $Q_*$  is bounded by an absolute constant. Since  $b_\ell(z) \leq 2e^{-z} \leq 2e^{-\theta z}$ , orthogonality gives

$$\beta_\ell(Q_*^2) \leq \frac{2}{\theta S} \leq C/d.$$

Lemma 3, uniformly also for this data-dependent polynomial, implies

$$\mu(K) \leq \mathbb{E}_\mu Q_*(x)^2 \leq \frac{C}{d} + C \left( \frac{d}{m} + e^{-cm} \right) \leq C/d.$$

Finally  $s_K = x_K/\alpha = O(1/d)$ . For every  $B$ ,  $A_B(K) = \prod_{j \in B} (1 - u_j(K)) \geq 1 - s_K \geq 1/2$  for large  $m$ . Hence

$$(\ell + 1)w_K/2 \leq \mu(K) \leq C/d, \quad w_K \leq Cm^{-3/2}.$$

Reversing the rows and replacing every  $p_j$  by  $1 - p_j$  gives the same bound for  $w_1$ . For a distinguished uniform die,  $w_q = 1/K$ ; the standard comparison tensor has  $m = n - 1$  and the stated mixed moments. This proves Theorem 1.